From the Softmax Jacobian to the AI Reallocation Derivative

A fully labeled derivation of $\frac{\partial \theta_k^*}{\partial G}$

1 Setup and notation

We allocate a fixed unit mass of labor across n sectors. Each sector s has a *logit* (a log-attractiveness score) z_s, and the *target share* θ_s^* is the softmax of those logits. Throughout, two index letters do two different jobs and must never be conflated: k is the *output* share we track, j is the *input* logit we perturb, and r is a throwaway summation dummy.

Symbol	Meaning	In the AI model
n	number of sectors	$\{a, m, \tau, k\}$, so $n = 4$
j	input index: the logit being perturbed	sector whose productivity shifts
k	output index: the share being tracked	usually KIS
r	dummy summation index	ranges over all sectors
z_s	logit of sector s	$z_s = \eta \ln p_s$
p_s	productivity of sector s	value added per worker
η	chase intensity (inverse temperature)	how hard labor chases productivity
e^{z_s}	unnormalized weight	p_s^η
S	partition function $\sum_r e^{z_r}$	shared normalizer $\sum_r p_r^\eta$
θ_s^*	target share e^{z_s}/S	target employment share
δ_{kj}	Kronecker delta (1 if $k{=}j$ else 0)	"same sector?" gate
g_s	net AI sensitivity $\alpha_s^G - \alpha_s^D \rho_s$	boost minus displacement
$\bar{g}$	weighted average $\sum_s \theta_s^* g_s$	economy-wide sensitivity

The object of the whole derivation is the response of one target share to the AI shock G:

$$\boxed{\frac{\partial \theta_k^*}{\partial G}} \qquad \text{where} \qquad \theta_k^* = \frac{e^{z_k}}{S}, \quad S = \sum_{r=1}^{n} e^{z_r}, \quad z_s = \eta \ln p_s.$$

We get there in three stages: (I) the pure-math Jacobian $\partial \theta_k^* / \partial z_j$; (II) the chain rule through $z_s = \eta \ln p_s$ and $p_s(G)$; (III) the relative-share reading that explains the empirical η_6 sign.

2 Stage I — the softmax Jacobian

Two reusable building blocks. The numerator e^{z_k} contains z_j *only* when $j = k$; the denominator S contains e^{z_j} *always*:

$$\frac{\partial}{\partial z_j} e^{z_k} = e^{z_k}\,\delta_{kj}, \qquad \frac{\partial S}{\partial z_j} = \frac{\partial}{\partial z_j}\sum_r e^{z_r} = e^{z_j}. \tag{1}$$

The second equality is the coupling channel: because S sits beneath every share, perturbing z_j feeds into all n outputs at once.

Quotient rule. With $u = e^{z_k}$, $v = S$, the rule $\partial(u/v) = (u'v - uv')/v^2$ and (1) give

$$\frac{\partial \theta_k^*}{\partial z_j} = \frac{(e^{z_k}\delta_{kj})\,S - e^{z_k}e^{z_j}}{S^2} = \delta_{kj}\underbrace{\frac{e^{z_k}}{S}}_{\theta_k^*} - \underbrace{\frac{e^{z_k}}{S}}_{\theta_k^*}\underbrace{\frac{e^{z_j}}{S}}_{\theta_j^*} = \theta_k^*(\delta_{kj} - \theta_j^*). \tag{2}$$

Log-derivative route (independent check). Since $\ln\theta_k^* = z_k - \ln S$, differentiating gives $\frac{1}{\theta_k^*}\partial_{z_j}\theta_k^* = \delta_{kj} - \frac{1}{S}\partial_{z_j}S = \delta_{kj} - \theta_j^*$, hence the same

Result.

$$\frac{\partial \theta_k^*}{\partial z_j} = \theta_k^*(\delta_{kj} - \theta_j^*) = \begin{cases} \theta_k^*(1-\theta_k^*) > 0 & j = k \quad (\textit{own leverage}) \\ -\theta_k^*\theta_j^* < 0 & j \neq k \quad (\textit{competition}). \end{cases}$$

In matrix form $J = \mathrm{diag}(\theta^*) - \theta^*\theta^{*\top}$, symmetric and singular ($J\mathbf{1} = 0$). Figure 1 shows the structure for $z = (2, 1, 0) \Rightarrow \theta^* = (0.665, 0.245, 0.090)$.

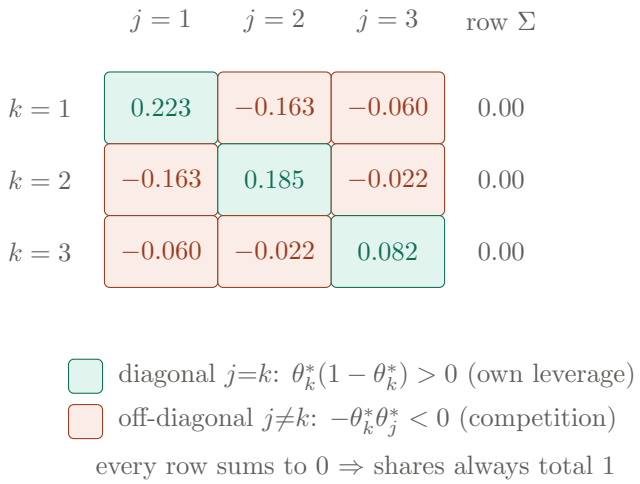


Figure 1: The softmax Jacobian $J_{kj} = \partial\theta_k^*/\partial z_j$. Row k is the share tracked; column j the logit nudged. Teal diagonal = a sector responding to its own shock; coral off-diagonal = mass stolen via the shared denominator S.

3 Stage II — chain rule to the AI shock

The logit depends on G through productivity, $z_s = \eta \ln p_s$ with $p_s = p_{s,t-1}\big[1 + \alpha_s^F F + \alpha_s^G G + \alpha_s^{FG} FG - \alpha_s^D \rho_s G\big]$. Differentiating and evaluating the bracket at the linearization point (≈ 1),

$$\frac{\partial z_s}{\partial G} = \eta \, \frac{\partial \ln p_s}{\partial G} \approx \eta\big(\alpha_s^G - \alpha_s^D \rho_s\big) = \eta \, g_s, \tag{3}$$

where $g_s \equiv \alpha_s^G - \alpha_s^D \rho_s$ is the *net AI sensitivity* of sector s (boost minus displacement drag). Figure 2 shows the schematic ordering: only KIS has its boost dominate its drag.

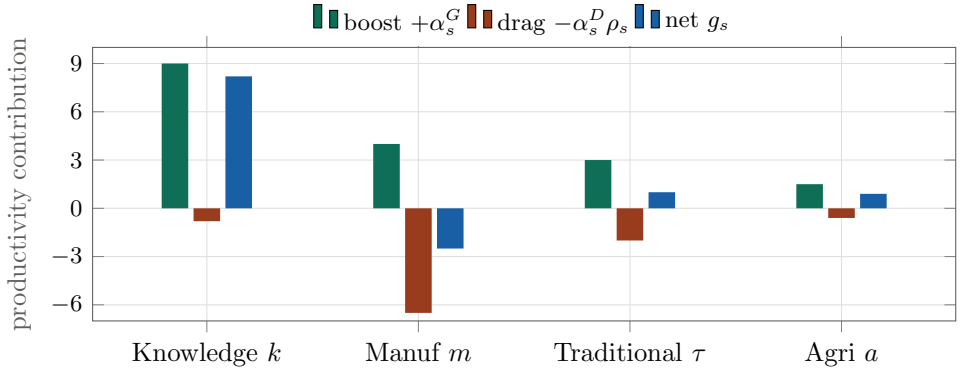


Figure 2: Net AI sensitivity $g_s = \alpha_s^G - \alpha_s^D \rho_s$ by sector (schematic). KIS is the only sector whose boost exceeds its displacement drag; manufacturing's net is negative. This ordering drives the reallocation.

Assembling the total derivative. Chain the Jacobian (2) through (3) over all logits:

$$\begin{aligned} \frac{\partial \theta_k^*}{\partial G} &= \sum_j \frac{\partial \theta_k^*}{\partial z_j} \frac{\partial z_j}{\partial G} = \sum_j \theta_k^* (\delta_{kj} - \theta_j^*) \, \eta g_j \\ &= \eta \, \theta_k^* \Big(\underbrace{\textstyle\sum_j \delta_{kj} g_j}_{= g_k} - \underbrace{\textstyle\sum_j \theta_j^* g_j}_{= \bar{g}} \Big) = \boxed{\eta \, \theta_k^* \big(g_k - \bar{g}\big)}. \end{aligned} \tag{4}$$

Result. *A sector's target share rises under AI* iff its net sensitivity beats the employment-weighted average*: $\partial \theta_k^* / \partial G > 0 \iff g_k > \bar{g}$. The competitor term θ_j^* from the Jacobian is exactly what converts raw sensitivities into the average $\bar{g}$.*

4 Stage III — the relative-share reading and the η_6 sign

Expand (4) for KIS ($\rho_k \approx 0 \Rightarrow g_k \approx \alpha_k^G$):

$$\frac{\partial \theta_k^*}{\partial G} \;\propto\; \eta \Big[\underbrace{\alpha_k^G}_{\text{KIS pull}} - \underbrace{\textstyle\sum_j \theta_j^* \alpha_j^G}_{\text{average boost}} + \underbrace{\textstyle\sum_j \theta_j^* \alpha_j^D \rho_j}_{\text{displacement elsewhere}} \Big]. \tag{5}$$

Note the displacement term enters with a + sign: routine displacement in *other* sectors lowers their shares and, through the shared S, mechanically *raises* KIS's relative share. Hence the two-sector object $\partial \theta_k^* / \partial G$ cannot be driven negative by displacement alone — a correction to the reduced-form expression $\eta \alpha_k^G - \eta \sum_{s \neq k} \theta_s^* \alpha_s^D \rho_s$, whose − sign mislocates the effect.

Where the negative legitimately lives: the informal sink τ. The premature-deindustrialization result is about *levels*—where displaced workers physically end up—not about KIS's relative share. With τ as an explicit third destination, the object that turns negative is the share reaching *productive* sectors:

$$\frac{\partial(\theta_k^* + \theta_m^*)}{\partial G} < 0 \quad \Longleftrightarrow \quad \frac{\partial \theta_\tau^*}{\partial G} > 0,$$

and by (4) $\partial\theta_\tau^*/\partial G > 0 \iff g_\tau > \bar{g}$. In a low-$H$, low-$\sigma^{\text{KIS}}$ economy the *realized* KIS boost collapses (α_k^G is gated by $\eta_6 H \sigma^{\text{KIS}}$, so the productivity gain has no base to land on), which drags $\bar{g}$ down below g_τ and sends labor into the sink. Figure 3 plots the productive-share derivative against absorptive capacity: it is the calculus behind the estimated $G \times \sigma^{\text{KIS}} < 0$ concentrated in lower-middle income.

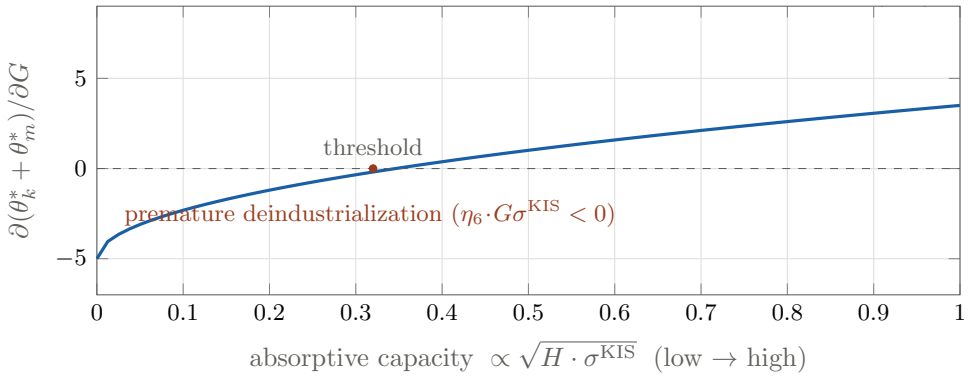


Figure 3: The sign flip. As human capital H and the KIS base σ^{KIS} rise, the realized KIS pull overtakes the routine push and the productive-share derivative crosses zero. LMICs sit left of the threshold ($\hat{\eta}_6 \approx -56$); UMICs sit near/right of it ($\hat{\eta}_6 \approx -1.7, \approx 0$).

Summary of the chain

$$\underbrace{\theta_k^*(\delta_{kj} - \theta_j^*)}_{\text{Jacobian (Stage I)}} \xrightarrow{\times \eta g_j,\ \sum_j} \underbrace{\eta\, \theta_k^*(g_k - \bar{g})}_{\text{AI derivative (Stage II)}} \xrightarrow{+\tau \text{ sink}} \underbrace{\frac{\partial(\theta_k^* + \theta_m^*)}{\partial G} \lessgtr 0}_{\text{premature deind. (Stage III)}}$$

The shared denominator S is the origin of inter-sector competition; the competitor term $-\theta_k^*\theta_j^*$ is that competition written as calculus; and the average $\bar{g}$ it produces is what makes "escape vs. trap" a contest between a sector's own AI sensitivity and the economy-wide average.